

Arthur Cancellieri Pires

Lead Data Scientist | Python, PyTorch | Anomaly Detection

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Professional Summary

Lead Data Scientist with 7+ years turning industrial sensor data into production ML for predictive maintenance. Expert in Python, PyTorch, and deep learning for time series and anomaly detection, delivering real-time failure predictions at 91% precision that cut unplanned downtime and maintenance costs. Led a 12-person team and shipped ML end to end: scalable pipelines, model deployment, and APIs (FastAPI, Docker). Ph.D. in mechanical engineering, grounding research rigor in real-world production.

Professional Experience

- Railway Laboratory (Lafer) – partnership with VALE** On-site
LEAD DATA SCIENTIST *Aug 2024 to Present*
 - Led a 12-person multidisciplinary team building end-to-end predictive maintenance ML systems for VALE, from stakeholder requirements to scalable production deployment.
 - Reduced equipment failure detection latency from three months to real time at 91% precision using deep learning models on multivariate sensor streams.
 - Developed unsupervised Transformer autoencoders that identified critical defects weeks before scheduled maintenance, including failure modes absent from the maintenance database.

Skills: Python | PyTorch | PyTorch Lightning | scikit-learn | Optuna | FastAPI | Docker | Data Science | Deep Learning | Transformers | Autoencoders | Anomaly Detection | Deployment
- Railway Laboratory (Lafer) – partnership with VALE** On-site
SENIOR DATA SCIENTIST *Jan 2022 to Aug 2024*
 - Designed an automated Python data-quality framework that scored the reliability of each data-collection run and filtered unreliable sensor data, replacing manual trip-by-trip screening.
 - Developed an unsupervised deep learning autoencoder that detected sensor anomalies missed by rule-based checks at about a 3% false-positive rate, preserving data quality for downstream modeling.
 - Established a systematic methodology in Python for characterizing baseline sensor behavior, improving the training and evaluation of downstream defect-detection models.

Skills: Python | PyTorch | scikit-learn | SQL | Optuna | Data Science | Deep Learning | Autoencoders | Anomaly Detection | Unsupervised Learning | Feature Engineering | Signal Processing | ETL | Data Quality
- Railway Laboratory (Lafer) – partnership with VALE** On-site
RESEARCH DATA SCIENTIST *Jun 2020 to Jan 2022*
 - Developed a Python ETL pipeline that processed 20,000–40,000 sensor records per asset per day, merging multiple raw data sources into a unified analytical dataset that powered all downstream machine learning models.
 - Built deep learning models that predicted asset condition with an R2 of approximately 0.98, reducing reliance on costly specialized inspections.
 - Created the data and modeling foundation in Python that supported subsequent predictive maintenance and track defect-detection initiatives at VALE.

Skills: Python | PyTorch | scikit-learn | SQL | Optuna | Data Science | Deep Learning | Feature Engineering | Signal Processing | ETL
- Tribology and Railway Dynamics Laboratory (LabTDF) – partnership with VALE** On-site
MACHINE LEARNING INTERN *Jan 2019 to Jun 2020*
 - Estimated hard-to-measure contact forces from onboard sensors using a simulation-validated ML pipeline, reaching an R2 of about 0.97 whose results helped drive VALE's decision to launch a dedicated R&D project on track predictive maintenance using instrumented wagons.
 - Optimized wheel and rail wear profiles using genetic algorithms, reducing wear by 20% and improving a key safety ratio by approximately 35% without sacrificing fatigue performance.
 - Delivered wear and fatigue analyses that drove adoption of a maintenance threshold change in operations, extending asset service life by 29%–50% (29% from the threshold change, up to 50% combined with the optimized wheel profile).

Skills: Python | scikit-learn | Optuna | MATLAB | Universal Mechanism | Machine Learning | Optimization | Genetic Algorithms | Multibody Simulation

Key Projects

- Industrial Sensor Anomaly Detection API** – github.com/pirao/AnomalyDetection2026
 - Production-style FastAPI service for real-time machine-fault detection that cut false positives to zero across 29 benchmark scenarios (F1 0.95, recall 0.91), up from a 0.28 F1 baseline.
 - Per-group residual scoring on multivariate vibration sensors with a stateful multi-window alert engine (confirmation and cooldown).
 - Serves versioned models from an MLflow registry. Containerized with Docker Compose and GitHub Actions CI.

Education

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- **State University of Campinas (UNICAMP)** Campinas, Brazil
PH.D. IN MECHANICAL ENGINEERING *Aug 2026*
 - **Dissertation:** A methodology for detecting track defects from instrumented railway vehicle data.
 - **State University of Campinas (UNICAMP)** Campinas, Brazil
M.SC. IN MECHANICAL ENGINEERING *Jan 2022*
 - **Thesis:** Measuring railway track irregularities from instrumented railway vehicle data using machine learning techniques.
 - **Federal University of Espirito Santo (UFES)** Vitoria, Brazil
B.SC. IN MECHANICAL ENGINEERING *May 2020*
 - **Thesis:** Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning.

Languages

Portuguese (Native), English (C1 - Fluent), Italian (A2 - Basic)

Publications & Conferences

SELECTED PUBLICATIONS

- 2026 **A practical approach for reliable track defect data from instrumented railway vehicles**, Heavy Rail
- 2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence
- 2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence
- 2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing

CONFERENCE PRESENTATIONS

- 2026 International Conference on Railway Technology (Railways 2026), Aug. 23–26 (accepted) *Budapest, Hungary*
- 2025 International Heavy Haul Association (IHHA) Conference, Nov. 17–21 *Colorado Springs, USA*
- 2023 International Heavy Haul Association (IHHA) Conference, Aug. 27–31 *Rio de Janeiro, Brazil*
- 2023 Railway Engineering Symposium, May 19–20 *Campinas, Brazil*